THE VIRTUAL MEET EXPERIENCE

2018 - 2019

HS VIRTUAL CHALLENGE MEET #2

COMPUTER SCIENCE

DO NOT OPEN TEST UNTIL TOLD TO DO SO

2018-2019 Virtual Challenge Meet #2 – Computer Science

Note: Correct responses are based on Java, **J2sdk v 1.8.xx**, from Sun Microsystems, Inc. All provided code segments are intended to be syntactically correct, unless otherwise stated (i. e. `error` is an answer choice) and any necessary Java 2 Standard Packages have been imported. Ignore any typographical errors and assume any undefined variables are defined as used. **For all output statements, assume that the System class has been statically imported…** ***import static java.lang.System.*;***

QUESTION 1

Which of the following is not equivalent to the expression $777_8 + 111111111_2$?

A. 1022_{10} B. 33232_4 C. 1776_8 D. $3fe_{16}$ E. 1111111110_2

QUESTION 2

What is output by the code to the right?

A. 84 B. 106

C. 212

D. There is no output due to a compile error.

E. There is no output due to a runtime error.

```
int x=212;
x = x / 2;
out.println(x);
```

QUESTION 3

What is output by the code to the right?

A. 7.0 B. 7.12

C. 7.2 D. 7.13

E. There is no output due to an error.

```
out.printf("%.2f", 7.123);
```

QUESTION 4

What is output by the code to the right?

A. 12 B. 11

C. 13 D. 10

E. -1

```
String s = "apluscompsci.com";
out.println( s.indexOf( "." ) );
```

QUESTION 5

What is output by the code to the right?

A. truefalse B. falsefalse

C. false D. true

E. There is no output due to a syntax error.

```
boolean a=true,b=false,c=true;
out.print(a&&b||c);
```

QUESTION 6

What is output by the code to the right?

A. 15 B. 12

C. 11 D. 14

E. There is no output due to an error.

```
Integer k = (int)Math.ceil(11.1);
out.println( k );
```

QUESTION 7

What is the output by the code to the right?

A. abc B. ab

C. ac D. bc

E. acb

```
int y=2;
switch(y) {
  case 1:out.print("a");
  case 2:out.print("b");
  default:out.println("c");
}
```

QUESTION 8

What is the output by the code to the right?

A. `3.14` B. `3.15`
C. `3` D. `3.00`
E. There is no output due to an error.

```
out.println( 3 + .14 );
```

QUESTION 9

What is output by the code to the right?

A. `0101` B. `1110`
C. `1011` D. `1010`
E. There is no output due to a syntax error.

```
int h=10;
String s=Integer.toBinaryString(h);
out.println(s);
```

QUESTION 10

What is the output by the code to the right ?

A. `6` B. `0`
C. `null`
D. There is no output due to a runtime error
E. There is no output due to a syntax error.

```
int[] mat[] = new int[3][4];
out.println(mat[2][3]);
```

QUESTION 11

What is the output by the code to the right?

A. `10`
B. `1`
C. `25`
D. There is no output due to a compile error.
E. There is no output due to a runtime error.

```
String s="5 6 -1";
int y=0;
Scanner sc=new Scanner(s);
while( sc.hasNextInt() ) {
  y += sc.nextInt();
}
out.println(y);
```

QUESTION 12

What is the output by the code to the right ?

A. `6` B. `10`
C. `8` D. `5`
E. `7`

```
int y=5;
for(int x=y; x< 10; x+=3)
  y++;
out.println(y);
```

QUESTION 13

What is the correct order of precedence for the operators to the right ?

A. `III, IV, II, I` B. `I, IV, III, II`
C. `II, IV, III, I` D. `I, II, III, IV`
E. `IV, III, II, I`

```
I. >=
II. >>
III. ==
IV. instanceof
```

QUESTION 14

What is the output by the code to the right ?

A. `4` B. `1`
C. `8` D. `2`
E. `6`

```
out.println( 5 - 2 + 5 );
```

QUESTION 15

What is the output by the code to the right ?

A. `[a, b, c]`

B. `[b, c]`

C. `[c]`

D. `[c, b, a]`

E. There is no output due to an error.

```
LinkedList<String>ll;
ll=new LinkedList<String>();
ll.add("a");
ll.add("b");
ll.add("c");
ll.remove();
ll.poll();
out.println(ll);
```

QUESTION 16

What is the output by the code to the right ?

A. `9` B. `15`

C. `10` D. `8`

E. There is no output due to an error.

```
out.println( 10 | 7 );
```

QUESTION 17

What is the output by the code to the right ?

A. `512` B. `2`

C. `8`

D. No output due to a syntax error

E. No output due to a runtime error.

```
BigInteger b=new BigInteger("512");
out.println(b);
```

QUESTION 18

What could replace **`<1*>`** in the code to the right, so that the numbers will be inserted into the stack?

A. `add` B. `push`

C. `put` D. A and B

E. None of the choices above is correct.

QUESTION 19

What could replace **`<2*>`** in the code to the right, so that each **`<2*>`** will remove a value from the stack?

A. `pop` B. `remove`

C. `pull` D. A and B

E. None of the choices above is correct.

```
Stack<Integer>s=new Stack<Integer>();
s. <1*> (4);
s. <1*> (6);
s. <1*> (14);
s. <2*> ();
s. <2*> ();
out.println(s);
```

QUESTION 20

Which of the following algorithms has the fastest worst-case run time?

A. binary search B. merge sort C. linear search D. insertion sort E. selection sort

QUESTION 21

What is the output by the code to the right ?

A. `[212, 17, 420]`

B. `[212, 17]`

C. `[212, 420]`

D. `[212, 420, 17]`

E. There is no output due to a runtime error.

```
ArrayList<String>a;
a=new ArrayList<String>();
a.add("212");
a.add("420");
a.add("17");
out.println(a);
```

QUESTION 22

Which of the following could replace **<1*>** in the code to the right?

A. `abstract`
B. `super`
C. `this`
D. `upper`
E. `Vehicle`

QUESTION 23

Assuming that **<1*>** is filled correctly, what is the output by `//1` ?

A. `Prius`
B. `Prius: 5`
C. `Prius: 5 mpg`
D. `5`
E. Output cannot be determined until runtime.

QUESTION 24

Assuming that **<1*>** is filled correctly, what is the output by `//2` ?

A. `Tracker: 4`
B. `Tracker: 4 mpg, 5 foot hull`
C. `Tracker: 4 mpg`
D. Output cannot be determined until runtime.
E. There is no output due to an error.

QUESTION 25

Assuming that **<1*>** is filled correctly, what is the output by `//3` ?

A. `false`
B. `true`
C. `Tracker: 4 mpg, 5 foot hull`
D. Output cannot be determined until runtime
E. This is no output due to a compile error.

```
class Vehicle{
  int mpg;
  String name;
  public Vehicle(int m,String n) {
     mpg=m;
     name=n;
  }
  public int getmpg() {
     return mpg;
  }
  public String toString() {
     return name+": "+mpg;
  }
}
class Boat extends Vehicle{
  int hull;
  public Boat(int m,int h,String n) {
     <1*>(m,n);
     hull=h;
  }
  public int getHull() {
     return hull;
  }
  public String toString() {
     return super.toString()+" mpg,
"+hull+" foot hull";
  }
}
/////////////client code/////////////
Vehicle v=new Vehicle(5,"Prius");
out.println(v);  //1
Vehicle b=new Boat(4,5,"Tracker");
out.println(b);  //2
out.println
     (v.getmpg()==b.getHull()); //3
```

QUESTION 26

What is output by the code to the right?

A. `falsetruefalse`
B. `falsetruetrue`
C. `falsefalsetrue`
D. `truetruetrue`
E. `truefalsetrue`

```
out.print("212wilcats!!!"
        .matches("(\\d\\w\\w!){3}"));
out.print
       ("Date_Mike".matches("\\w*"));
out.print("TheBigCheese".matches
       ("([A-Z][a-z][a-z]){3}\\w+"));
```

QUESTION 27

Which of the following implements the `List` interface?

A. `HashSet`

B. `LinkedList`

C. `TreeSet`

D. `Queue`

E. All of the above implement `List`.

QUESTION 28

What is output by the line marked `//q28`?

A. `4` B. `8`

C. `32` D. `16`

E. `0`

QUESTION 29

What is output by the line marked `//q29`?

A. `64` B. `128`

C. `256` D. `512`

E. `0`

QUESTION 30

What is output by the line marked `//q30`?

A. `4096` B. `2048`

C. `512` D. `1024`

E. `0`

```
public static int recur(int r) {
 if(r<=0)
  return 1;
 return recur(r-1)*recur(r-2)*2;
}
///////client code//////////////////
out.println(recur(3));  //q28
out.println(recur(4));  //q29
out.println(recur(5));  //q30
```

QUESTION 31

Which of the following data structures operates in a last in first out(LIFO) manner?

A. Queue B. Stack C. List D. Map E.HashTable

QUESTION 32

What is output by the line labeled `//1`?

A. `212` B. `420`

C. `8` D. `4`

E. None of the above lines cause an error.

QUESTION 33

What is output by the line labeled `//2`?

A. `[8, 9, 8, 67, 212, 17, 420]`

B. `[8, 8, 9, 17, 67, 212, 420]`

C. `[212, 67, 17, 9, 8, 8, 4]`

D. `[212, 67, 9, 17, 8, 8, 4]`

E. There is no output due to a compile error.

```
PriorityQueue<Integer> q;
q=new PriorityQueue<Integer>();
q.add(9);
q.add(8);
q.add(17);
q.add(4);
q.add(212);
q.add(8);
q.add(420);
q.add(67);
out.println(q.remove()); //1
out.println(q); //2
```

QUESTION 34

What could replace **<1*>** in the code to the right?

A. `implements Comparable`
B. `implements Comparable<Pair>`
C. `implements Comparable<E>`
D. More than one of the above.
E. None of the above.

QUESTION 35

What is outptut by the line marked `//1` in the client code to the right?

A. `null`
B. `[]`
C. `0`
D. `[, , , , , , , , , , ]`
E. There is no output due to an error.

QUESTION 36

What is output by the line marked `//2` in the client code to the right ?

A. `SlimShady`
B. `funzo`
C. `5 > 7`
D. `3`
E. There is no output due to an error.

QUESTION 37

What is output by the line marked `//3` in the client code to the right ?

A. `[2, 3, 4, 6]`
B. `[5 > 7, 3, 420, 212]`
C. `[0, 2, 3, 4, 6]`
D. `[SlimShady, 5 > 7, 3, 420, 212]`
E. There is no output due to a compile error.

QUESTION 38

What data structure is demonstrated by the code to the right ?

A. a set
B. a priority queue
C. a map
D. a queue
E. a hashtable

```
class Structure{
  ArrayList<Pair>a;
  public Structure() {
          a=new ArrayList<Pair>();
  }
  public void add(int y,String s) {
          a.add(new Pair(y,s));
          Collections.sort(a);
  }
  public String remove() {
          return a.remove(0).str;
  }
  public String toString() {
          return ""+a;
  }
  public String peek() {
          return a.get(0).str;
  }
}
class Pair <1*>{
  int in;
  String str;
  public Pair(int i,String s) {
          in=i;str=s;
  }
  public int compareTo(Pair p) {
          return in-p.in;
  }
  public String toString() {
          return str;
  }
}

//////////////client code///////////
Structure s=new Structure();
out.println(s.a);  //1
s.add(3, "3");
s.add(4, "420");
s.add(6, "212");
s.add(-1, "funzo");
s.add(2, "5 > 7");
s.add(0, "SlimShady");
s.remove();
out.println(s.peek());  //2
out.println(s);  //3
```

QUESTION 39

What is the most simplified form of the following boolean algebra expression?

```
A*B+0*B+(B*C*A)+0*A
```

QUESTION 40

What is the two's complement 8-bit binary value of the following base 10 value ?

```
-54
```

Name _____ Final Score _____ _____

School _____ 2nd Grading _____ _____

Division: 1A 2A 3A 4A 5A 6A Grade: 9 10 11 12 1st Grading _____ _____
score initials

The Virtual Challenge Meets
Computer Science - Student Answer Sheet

1)	_____	21)	_____
2)	_____	22)	_____
3)	_____	23)	_____
4)	_____	24)	_____
5)	_____	25)	_____
6)	_____	26)	_____
7)	_____	27)	_____
8)	_____	28)	_____
9)	_____	29)	_____
10)	_____	30)	_____
11)	_____	31)	_____
12)	_____	32)	_____
13)	_____	33)	_____
14)	_____	34)	_____
15)	_____	35)	_____
16)	_____	36)	_____
17)	_____	37)	_____
18)	_____	38)	_____
19)	_____	39)	_____
20)	_____	40)	_____

Total Number Correct _____ x 6 = _____

Total Number Incorrect _____ x -2 = _____

Final Score (record here and top right) _____

2018-2019 Virtual Challenge Meet #2
Computer Science - KEY

1) B
2) B
3) B
4) A
5) D
6) B
7) D
8) A
9) D
10) B

11) A
12) E
13) C
14) C
15) C
16) B
17) A
18) D
19) A
20) A

21) D
22) B
23) B
24) B
25) E
26) B
27) B
28) D
29) B
30) A

31) B
32) D
33) A
34) B
35) B
36) A
37) D
38) B
39) A*B or A&&B
40) 11001010

Note to Graders:

- All provided code segments are intended to be syntactically correct, unless otherwise stated (e.g. error is an answer). **Ignore any typographical errors**.
- Any necessary Standard Java 2 Packages are assumed to have been imported as needed.
- Assume any undefined (undeclared) variables have been defined as used.

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